

HW #4 (Chapter 4)

~~4.16~~ 如圖 4.37 中機構，若 $\omega_2 = 5 \text{ rad/s}$ ，順時針方向，試求 $\vec{V}_D$ 及 ω_3 。

4.11

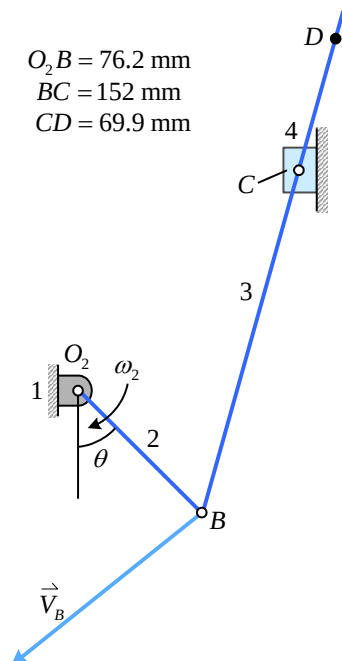


圖 4.37

解：(1) $V_B = (O_2B) \omega_2$
 $= 0.0762 \times 5$
 $= 0.381 \text{ m/s}$

速度多邊形以

$1 \text{ mm} = 0.006 \text{ m/s}$

(2) $\vec{V}_C = \vec{V}_B + \vec{V}_{C/B}$ 可得 C' 點

(3) 由 $BD = \frac{220}{152} BC = 1.46 BC$

知 $B'D' = 1.46 B'C'$

可得 D' 點

(4) 按比例算出

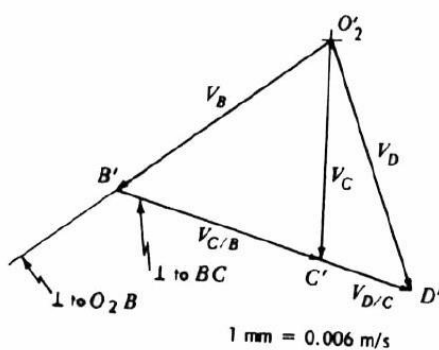
$V_D = 0.387 \text{ m/s}$

$V_{D/B} = 0.457 \text{ m/s}$

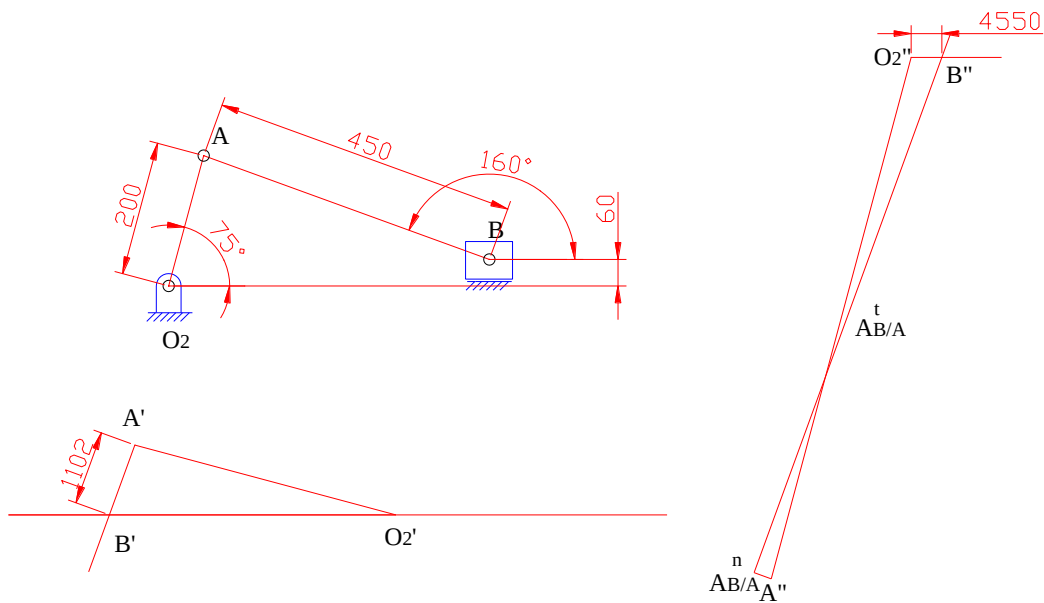
(5) $\omega_3 = \frac{V_{C/B}}{BC} = \frac{V_{D/B}}{BD}$

即 $\omega_3 = \frac{0.457}{0.222}$

$= 0.06 \text{ rad/s (ccw)}$



4.13



$$\begin{aligned} \vec{V}_B &= \vec{V}_A + \vec{V}_{B/A} \\ ? \quad O_2 A \times \omega_2 = 200 \times 20 = 4000 \text{ mm/s} & \quad AB \times \omega_3 \\ \angle 0^\circ \quad \perp l_2 & \quad \perp l_3 \end{aligned}$$

$$\therefore V_{B/A} = 1102 \text{ mm/s}$$

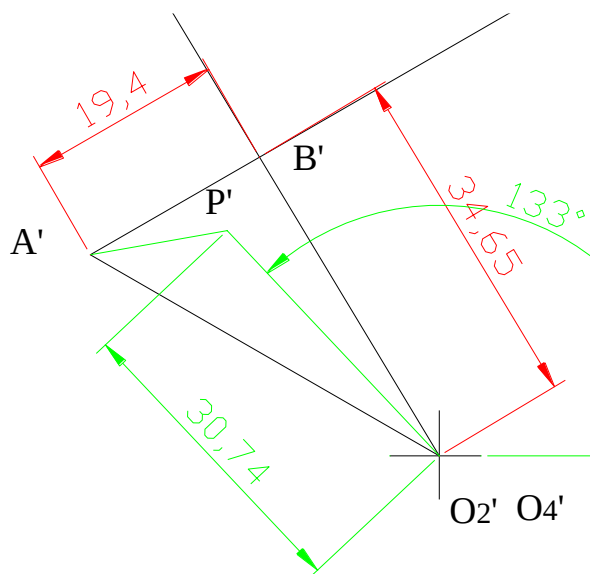
$$\begin{aligned} \vec{A}_B &= \vec{A}_A^n + \vec{A}_A^t + \vec{A}_{B/A}^n + \vec{A}_{B/A}^t \\ ? \quad O_2 A \times \omega_2^2 = 200 \times 20^2 = 80000 \text{ mm/s}^2 & \quad 0 \quad V_{B/A}^2 / AB = 1102^2 / 450 = 2699 \text{ mm/s}^2 \\ \angle 0^\circ \quad // l_2, A \rightarrow O_2 & \quad // l_3, B \rightarrow A \quad \perp l_3 \end{aligned}$$

$$\therefore \text{滑塊之加速度為 } 4500 \text{ mm/s}^2, \angle 0^\circ$$

4.20

解：

4.14



$$\begin{array}{ccccc} \overset{-V}{V}_B & = & \overset{V}{V}_A & + & \overset{-V}{V}_{B/A} \\ ? & & 1200 \text{ mm/s} & & ? \\ \angle 121^\circ & & \perp l_2 = 150^\circ & & \perp l_3 = 30^\circ \end{array}$$

$$\omega_2 = 600 \cdot 2\pi / 60 = 20\pi \text{ rad/s}$$

$$V_A = 60 \cdot 20\pi = 1200\pi \text{ mm/s}$$

$$\text{取 } 1\text{mm} = 30\pi \text{ mm/s 繪圖 } V_A = 40\text{mm}$$

$$\text{由圖面得 } V_B = 34.65\text{mm 長} \Rightarrow V_B = 3266\text{mm/s},$$

$$\omega_4 = V_{B/O_4B} = -3266/100 = -32.66\text{rad/s}$$

$$V_{B/A} = 19.4\text{mm 長} \Rightarrow V_{B/A} = 1828\text{mm/s},$$

$$\omega_3 = V_{B/A}/AB = 1828/120 = 15.23\text{rad/s}$$

$$\begin{array}{ccccc} \overset{V}{V}_P & = & \overset{V}{V}_A & + & \overset{V}{V}_{P/A} \\ ? & & 1200 \text{ mm/s} & & 15.23 \cdot 85 \\ ? & & \perp l_2 = 150^\circ & & \perp AP = 10^\circ \end{array}$$

$$V_P = 30.74 \cdot 30\pi \text{ mm/s} = 2897\text{mm/s} < 133^\circ$$

$$\begin{array}{ccccccc} \overset{V}{A}_B^n & + & \overset{-V}{A}_B^t & = & \overset{V}{A}_A^n & + & \overset{V}{A}_A^t & + & \overset{V}{A}_{B/A}^n & + & \overset{-V}{A}_{B/A}^t \\ O_4B \times \omega_4^2 & ? & O_2A \times \omega_2^2 & & & & V_{B/A}^2/AB & & & & \\ = 100 \cdot 32.66^2 & & = 60 \cdot 10^2 & & 0 & & = 1828^2/120 & & ? & & \text{依比例繪} \\ = 106667\text{mm/s}^2 & & = 6000\text{mm/s}^2 & & & & = 27846\text{mm/s}^2 & & & & \\ \angle 31^\circ & \perp l_4 = 121^\circ & // l_2, A \rightarrow O_2 = 240^\circ & & & & // l_3, B \rightarrow A = 120^\circ & & \perp l_3 = 210^\circ & & \end{array}$$

圖即可求解。